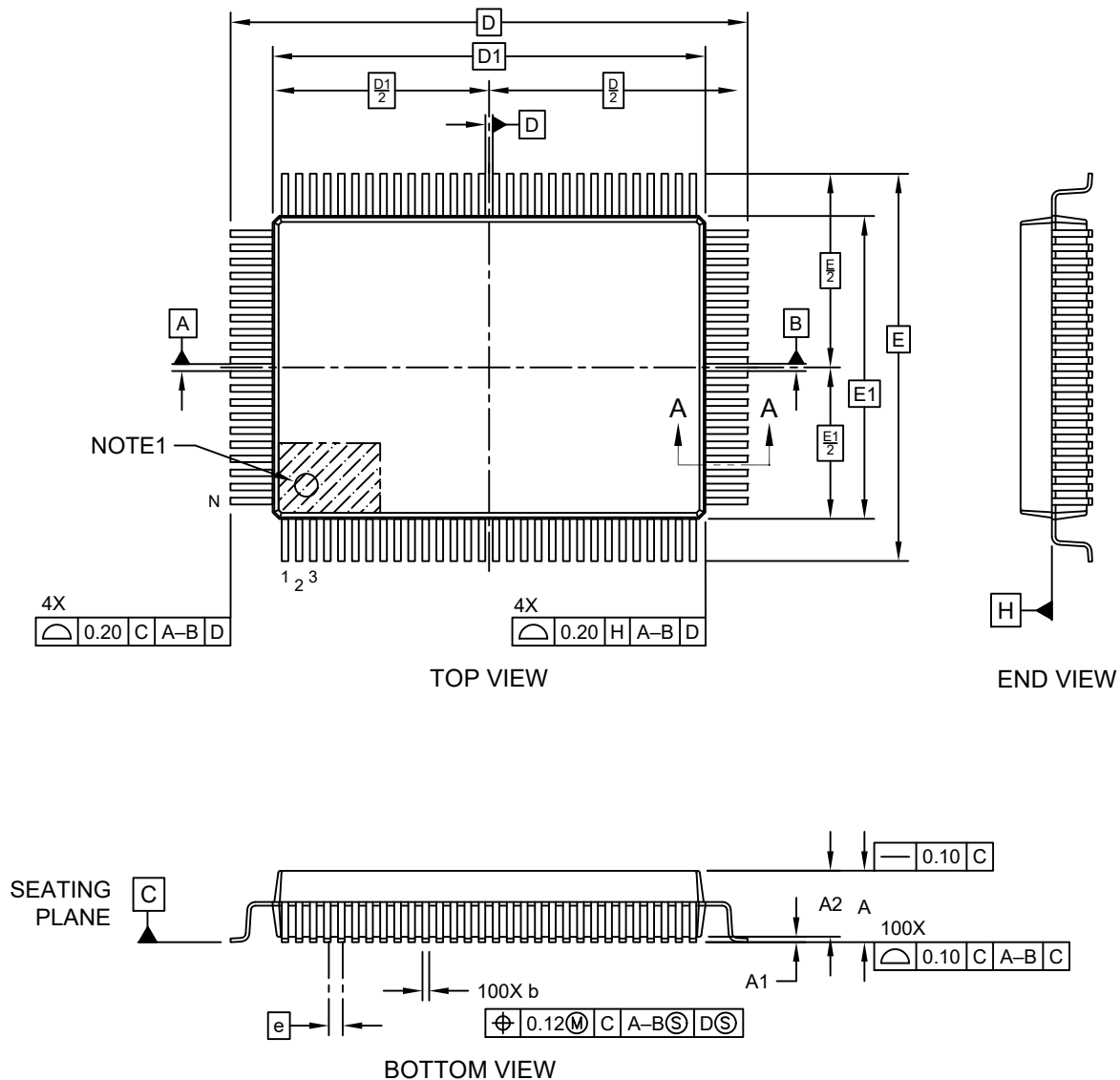


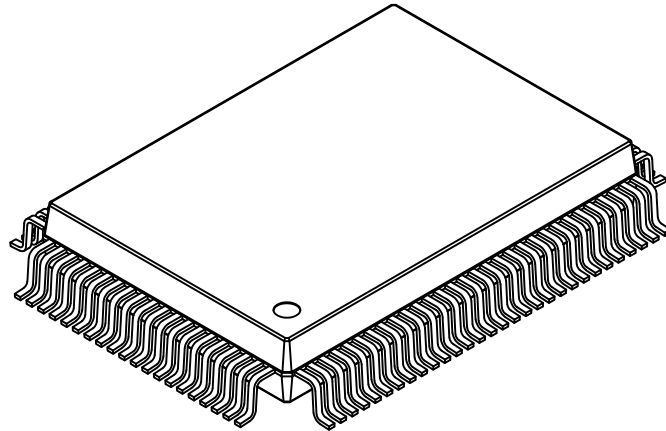
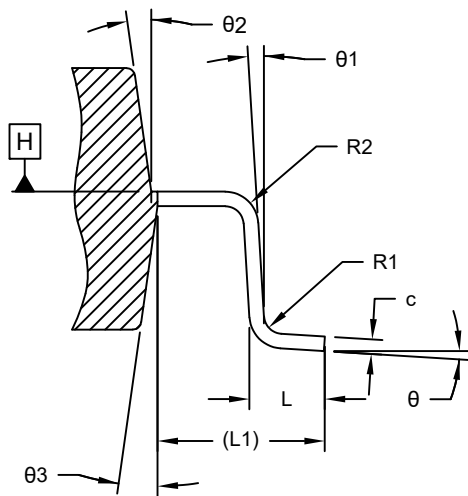
100-Lead Plastic Quad Flat Pack (KKB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



100-Lead Plastic Quad Flat Pack (KKB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



SECTION A-A

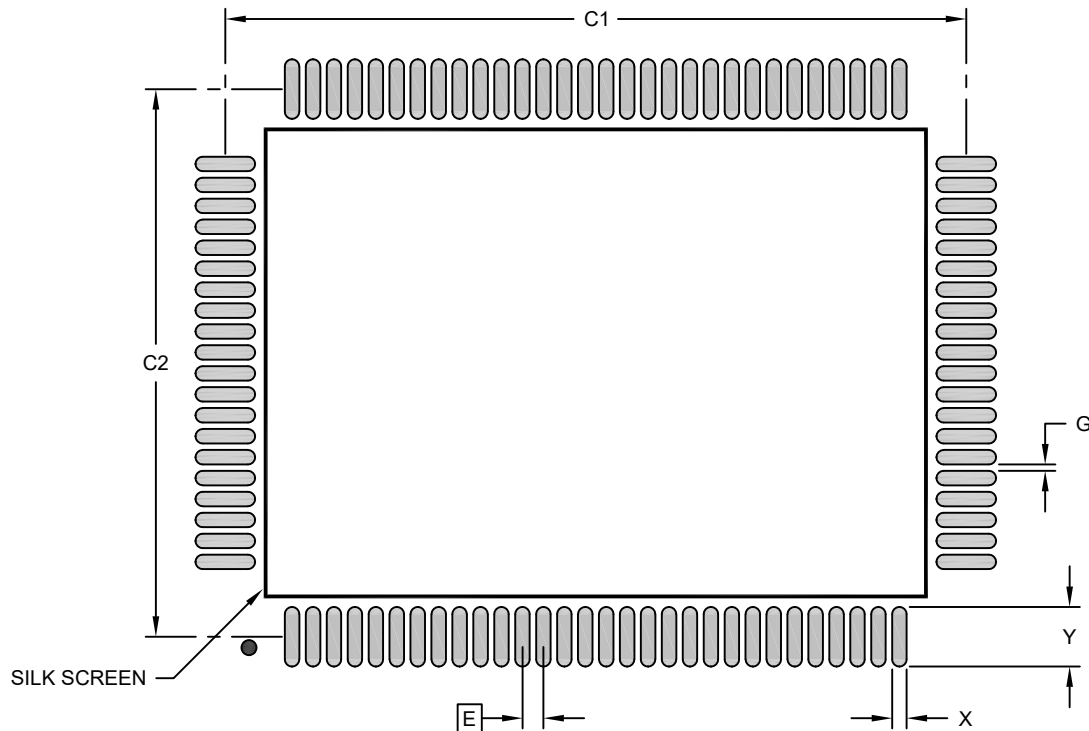
Dimension	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.65 BSC		
Overall Height	A	0.00	–	3.30
Standoff	A1	–	–	0.25
Molded Package Thickness	A2	2.55	2.80	3.05
Overall Length	D	23.90 BSC		
Molded Package Length	D1	20.00 BSC		
Overall Width	E	17.90 BSC		
Molded Package Width	E1	14.00 BSC		
Terminal Width	b	0.22	–	0.38
Terminal Thickness	c	0.11	–	0.23
Terminal Length	L	0.73	0.88	1.03
Footprint	L1	1.95 REF		
Lead Bend Radius	R1	0.13	–	0.30
Lead Bend Radius	R2	0.13	–	–
Foot Angle	θ	0°	3.5°	7°
Lead Angle	θ1	0°	–	–
Mold Draft Angle	θ2	12°	–	14°
Mold Draft Angle	θ3	12°	–	14°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

100-Lead Plastic Quad Flat Pack (KKB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1		23.00	
Contact Pad Spacing	C2		17.00	
Contact Pad Width (X100)	X			0.45
Contact Pad Length (X100)	Y			1.85
Contact Pad to Contact Pad (X96)	G	0.20		

Notes:

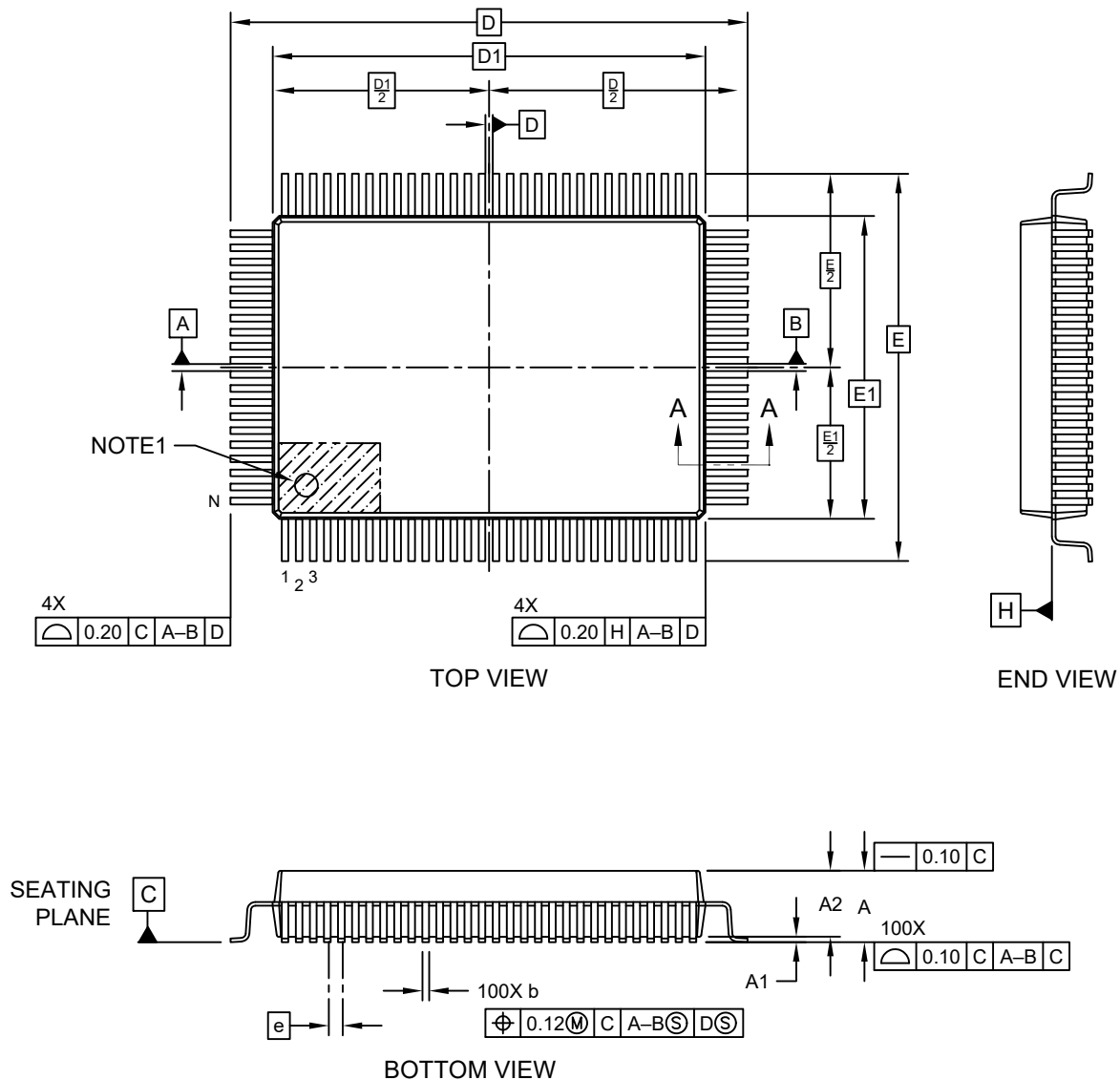
1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23293-KKB Rev B

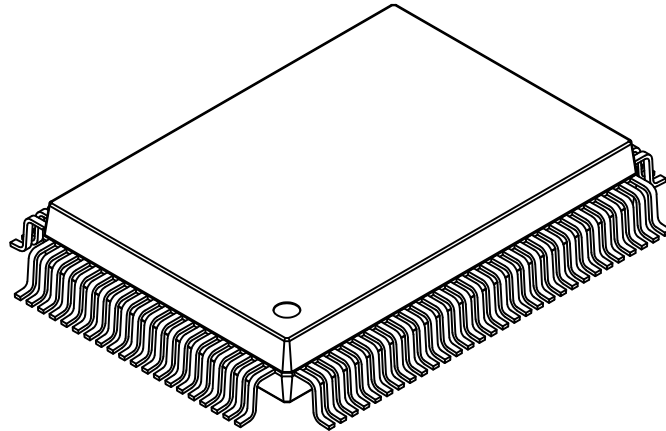
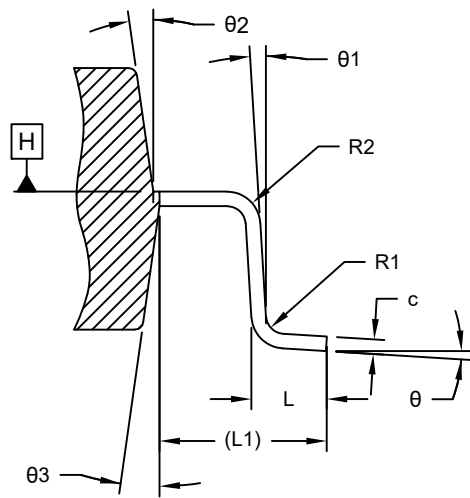
100-Lead Plastic Quad Flat Pack (KLB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



100-Lead Plastic Quad Flat Pack (KLB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



SECTION A-A

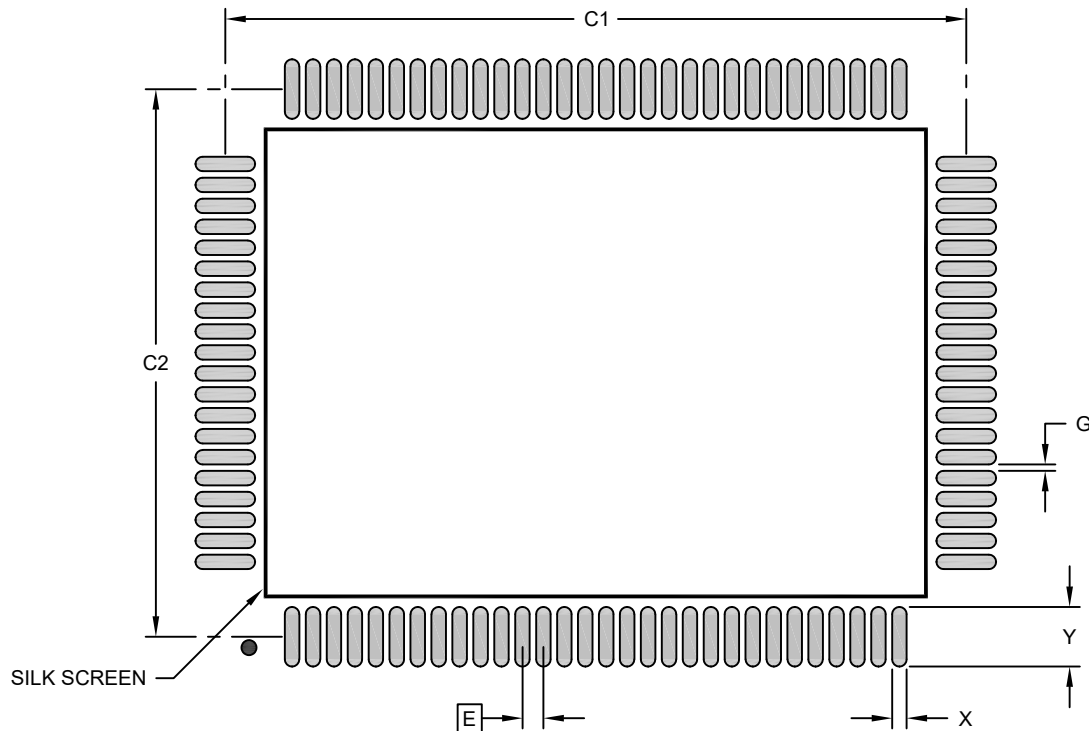
Dimension	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	100		
Pitch	e	0.65 BSC		
Overall Height	A	0.00	—	3.30
Standoff	A1	—	—	0.25
Molded Package Thickness	A2	2.55	2.80	3.05
Overall Length	D	23.90 BSC		
Molded Package Length	D1	20.00 BSC		
Overall Width	E	17.90 BSC		
Molded Package Width	E1	14.00 BSC		
Terminal Width	b	0.22	—	0.38
Terminal Thickness	c	0.11	—	0.23
Terminal Length	L	0.73	0.88	1.03
Footprint	L1	1.95 REF		
Lead Bend Radius	R1	0.13	—	0.30
Lead Bend Radius	R2	0.13	—	—
Foot Angle	θ	0°	3.5°	7°
Lead Angle	θ1	0°	—	—
Mold Draft Angle	θ2	12°	—	14°
Mold Draft Angle	θ3	12°	—	14°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

100-Lead Plastic Quad Flat Pack (KLB) - 20x14x2.80 mm Body [PQFP] Atmel Legacy Package QTC

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.65 BSC		
Contact Pad Spacing	C1		23.00	
Contact Pad Spacing	C2		17.00	
Contact Pad Width (X100)	X			0.45
Contact Pad Length (X100)	Y			1.85
Contact Pad to Contact Pad (X96)	G	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23293-KLB Rev B